



Full Length Article
Infectious Disease

Steroid-Sensitive, but Not Steroid-Dependent or Steroid-Resistant Acute Graft-versus-Host Disease, Results in Similar Infection Risk as No Graft-versus-Host Disease following Allogeneic Hematopoietic Cell Transplantation



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A B S T R A C T

Patients with acute graft-versus-host disease (GVHD) have an increased risk for infectious complications after allogeneic hematopoietic cell transplantation (HCT), but the risk according to response to therapy has not been well studied. We performed a retrospective analysis of the infectious complications for 1 year following allogeneic HCT at the University of Minnesota including 1143 pediatric and adult patients with and without aGVHD. The patients with aGVHD were classified into treatment response groups based on response to corticosteroids as first-line therapy: steroid-sensitive (SS; n = 114), steroid-resistant (SR; n = 103), and steroid-dependent (SD; n = 168) aGVHD. We observed that the cumulative incidence and density of infections in patients with SS aGVHD parallel the values in patients without GVHD. Infection density (ie, number of infections occurring per 100 days at risk) was greater in the patients with aGVHD compared with patients in both early and later post-transplantation periods. In GVHD patients, among the infections developed from the onset of aGVHD through 80 days of treatment, and until 1 year following transplantation, SS and SD patients had fewer bacterial and viral infections than SR patients. The overlap of nonrelapse mortality between SS and SD GVHD patients is a function of SD GVHD being responsive to steroid therapy, even if continued therapy is required. In summary, although valid goals may include reducing unneeded antibacterial antibiotic therapy and preserving microbiome diversity, these data suggest that anti-infective therapy is justified by the density of infections observed during active GVHD treatment.

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INTRODUCTION

Acute graft-versus-host disease (aGVHD) remains the major cause of nonrelapse mortality (NRM) and morbidity after allogeneic hematopoietic cell transplantation (HCT). Although immunosuppression may effectively prevent and treat aGVHD, these therapies, as well as aGVHD itself, may render patients more vulnerable to infectious complications [1–9]. The incidence and nature of opportunistic infections in patients treated for aGVHD have been incompletely defined.

Studies of specific steroid-refractory (SR) aGVHD treatment approaches report infections among treated patients but without comparison to other categories of steroid responsiveness [10–20].

In our previous comparison of steroid-sensitive (SS), steroid-dependent (SD), and SR aGVHD treatment cohorts [21], the SS and SD aGVHD groups had similar risks of overall mortality and NRM. SD aGVHD did not have an intermediate prognosis between the SS and SR groups; only the development of SR aGVHD led to greater NRM and overall mortality. In an effort to provide additional data regarding the impact and morbidity of infections that developed among patients in these groups, here we report a detailed analysis of the infectious complications for 1 year following transplantation for consecutive GVHD and control non-GVHD patients [21].

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METHODS

Cohort Synopsis

The study population included 1143 consecutive adult and pediatric allogeneic HCT recipients from the University of Minnesota who underwent HCT between 2008 and 2016. Only first allogeneic transplantations were included. All patients or their guardians provided written informed consent allowing the use of their medical data in clinical research analysis. The University of Minnesota Institutional Review Board reviewed and approved this study.

Infection prophylaxis was based on written local institutional guidelines that include electronic medical record order sets protocols based on national guidelines [22]. Individual patient antimicrobial treatment likely varied from guidelines in settings of patient medication allergies and/or organ function. From day -1, the standard prophylactic antibiotic for both adults and children was levofloxacin, which was continued until the first day of absolute neutrophil count >1000 cells/ μ L, unless the patient required escalated antibiotics for neutropenic fever. Levofloxacin (for pneumococcal prophylaxis) is restarted for patients with steroid use or diagnosed with GVHD and continued until 1 to 2 months off immunosuppressive medications.

Patients underwent the transplantation procedure in rooms ventilated with high-efficiency particulate air filtration systems to prevent acquisition of exogenous mold infections [23]. At a minimum, patients received azole-based yeast prophylaxis during the first month post-transplantation, with patients considered high risk for early mold infection receiving mold prophylaxis. The standard prophylactic antifungal agent was fluconazole, unless there was a reason for an individual patient to be on mold-active azole therapy (ie, history of fungal infection, suspicious pulmonary nodules, history of current or prolonged functional or actual neutropenia). The inpatient HCT order set prepopulates with voriconazole as the mold-active azole, although a different azole such as posaconazole is substituted if the patient had received posaconazole prior to admission or did not tolerate voriconazole after it was started. Therapeutic drug monitoring is provided for patients on mold-active azole therapy. Fungal prophylaxis is continued through an immunosuppression taper to avoid large swings in calcineurin inhibitor levels.

During periods of neutropenia, broad-spectrum antibacterial agents were administered that included antipseudomonal antibacterial agents for febrile neutropenia [24]. During periods of fever, assessment for infection and augmentation of antimicrobial agents included empirical treatment for yeasts and molds for persistent fever.

Acyclovir was used to prevent herpesvirus (including varicella) infections. This study predated availability of letermovir, so the standard prophylactic viral agent was high-dose acyclovir (i.v. 3 times per day or oral 5 times per day) through day+100, then oral twice daily through 1 year. Monitoring guidelines were followed to detect and preemptively treat cytomegalovirus infection. Acyclovir was restarted for patients with steroid use or diagnosed with GVHD and continued until tapered off immunosuppressive medications.

Medication to prevent *Pneumocystis* infections was started within 1 month post-transplantation, following engraftment. The standard agent was trimethoprim/sulfamethoxazole, which was continued through 1 year or until discontinuation of GVHD treatment.

For patients diagnosed with aGVHD, initial therapy consisted of oral prednisone at approximately 2 mg/kg/day for adults or 60 mg/m²/day for children (or the i.v. equivalent of methylprednisolone). Initial treatment dosing was divided dose 3 times daily for the first 7 days and then a single dose for 7 days, with subsequent scheduled tapering over 8 weeks.

Grading of steroid response followed standard guidelines [25,26], and flowed into a final categorization of steroid response following a Sankey flow principle [27]. SS aGVHD cases had successful tapering off steroids by day 80 of aGVHD treatment and no later flare of GVHD. SR aGVHD cases had new or progressive aGVHD while on therapy, or death before day 56 of GVHD treatment. SD aGVHD required steroids beyond day 80 of aGVHD treatment due to limited response or flare during taper. Further detail regarding steroid response category (SS, SR, and SD) aGVHD definitions have been previously reported [21]. For aGVHD patients requiring immunosuppressive treatment, bacterial, viral, fungal, and *Pneumocystis* prophylaxes were continued for the duration of immunosuppression.

Data Collection and Audit

Infection episodes were recorded from the day of transplantation to death, graft failure, relapse, second transplantation, or until 1 year post-HCT. Data from the first 100 days from HCT were collected from laboratory reports, and data for the remainder of the follow-up period were collected from medical provider notes, including outside medical provider correspondence.

The start date of an infection episode was defined as the date on which any infection was confirmed by culture, histology, cytology, polymerase chain reaction, or antigenemia. Infections that had the potential to be interpreted as contaminants (eg, coagulase-negative *Staphylococcus* in blood culture) were counted as infection if treatment was initiated. Dermatomal varicella zoster reactivation did not require microbiologic confirmation to be documented as an infectious episode. Cases of probable invasive fungal

infection based on cytology findings of unsuspected fungal elements with compatible clinical picture were retained.

Clinically compatible time frames were used to define one infectious episode from a second episode with the same organism. For CMV and herpes simplex virus, the maximum allowed interval for a single infectious event was 60 days, whereas for herpes zoster virus and other viral infections, it was 14 days. For all bacteria other than *Clostridioides difficile*, the allowed interval was 7 days; for *C. difficile*, it was 31 days. The allowed intervals were 14 days for yeast infections (*Candida* and *Cryptococcus* spp.), and 90 days for molds (*Aspergillus*, *Fusarium*, and *Mucor* spp.).

Based on previous studies showing the importance of auditing infection data [28,29], a convenience random sample of 175 patients was audited to check for completeness of infection data capture. The auditor received an infection summary for each patient listing the date, organism, body site, and comments regarding each infection as prospectively reported to the database. Each auditor confirmed database information by reviewing electronically generated medical records of microbiology, virology, imaging, and surgical pathology reports supplemented by clinical notes. The audit sample found and corrected minor discrepancies for the infection dataset but no major discrepancies, so a full audit of the remaining 968 patients was not conducted.

Statistical Analysis

Infectious complications were analyzed through comparisons of patients with and without aGVHD and among the 3 GVHD steroid response categories. Data on time to infection or other events were calculated from the date of transplantation. Patients were censored at relapse, second HCT, 1 year post-transplantation, death, or last follow-up, whichever came first. When a single organism was recovered multiple times, we followed an algorithm for infection recurrence intervals (as above) to define infection episodes; these were specific to microorganisms and have been reported previously [28,29]. The number of patients with infections reported was summarized for each infection type and by aGVHD cohort.

Analysis of infection incidence, by frequency and cumulative incidence (CI), and infection density (number of infections per 100 days for patients with repeated infections) was performed for overall infections, bacterial infections, viral infections and fungal infections, over different time periods. Days at risk for infection was adjusted for days alive within the specified period.

Time periods for the first analysis of infection density mimicked the phases of predictable opportunistic infections experienced by all HCT recipients: day 0 to day 30, day 31 to day 100, day 101 to 1 year, and overall within 1 year of HCT. Time periods for the second analysis were based on landmarks experienced by the individual GVHD patients on this study: day 0 to the day prior to engraftment, the date of engraftment to the day prior to the start of aGVHD steroid therapy, the date of aGVHD therapy to 80 days later (when steroid tapering was complete for SS GVHD patients), and day 81 of aGVHD therapy to 1 year after HCT.

Death was considered a competing risk in CI analysis. We used the log-rank test to compare distributions of CI. Infection density was computed by infection frequencies over 100 patient-days at risk. We compared infection density between aGVHD patients and those without GVHD using the Mantel-Haenszel chi-square test.

RESULTS

Study Population

A total of 1143 patients were studied and of these, 385 patients (34%) developed aGVHD requiring systemic corticosteroid therapy; the remaining 758 patients without GVHD were considered controls (66%). There were some demographic distinctions between patients with and without aGVHD (Table 1; additional details provided in [5]). More patients in the aGVHD cohort were age >18 years compared with the control group (83% versus 49%), and more had a low (12% versus 6%) or intermediate (63% versus 38%) Disease Risk Index (DRI; validated only for adult HCT recipients). More patients in the control group had a nonmalignant diagnosis as the reason for transplantation (42% versus 10%). The most frequently used GVHD prophylaxis regimen in both groups was cyclosporine plus mycophenolate mofetil (75% in the aGVHD group and 63% in controls).

GVHD occurred in 16% of haploidentical transplantations, 31% of both sibling and unrelated donor transplantations, and 39% of cord blood transplantations. Of the 385 patients with aGVHD, 114 (30%) had SS aGVHD, 103 (27%) had SD aGVHD, and 168 (43%) had SR aGVHD. SR GVHD was observed in 6.6% of

Table 1

Patients with Infections in the First Year Post-Transplantation, by aGVHD and Steroid Response Subgroups

Factor	Frequency					P Value
	All aGVHD	aGVHD Treated with Systemic Steroids			No GVHD	
		SS (30%)	SD (27%)	SR (43%)		
Patients per group	385	114	103	168	758	
Male sex, n (%)	242 (63)	71 (62)	65 (63)	106 (63)	458 (60)	.88
Age						<.01
Median (range)	50 (29-60)	46 (0-73)	49 (2-73)	51 (0-75)	17 (0-74)	
<18 yr, n (%)	67 (17)				390 (51)	
Donor type, n (column %)						<.01
Sibling match	112 (29.1)	31 (27.2)	35 (34)	46 (27.4)	249 (32.9)	
Haploidentical	10 (2.6)	3 (2.6)	3 (3)	4 (2.4)	51 (6.7)	
Unrelated	60 (15.6)	11 (9.7)	10 (10)	39 (23.2)	135 (17.8)	
Allele match	52	8	9	35	109	
Partial match	6	2	0	4	26	
Mismatch	2	1	1	0	0	
Cord blood	203 (52.7)	69 (60.5)	55 (53)	79 (47)	323 (42.6)	
Single	55	28	13	14	158	
Double	148	41	42	65	165	
Donor type, n (% of donor type)						<.01
Sibling matched (N = 361)	112 (31)	31 (9)	35 (10)	46 (13)	249 (69)	
Haploidentical (N = 61)	10 (16)	3 (5)	3 (5)	4 (6.6)	51 (84)	
Unrelated (N = 195)	60 (31)	11 (2)	10 (5)	39 (20)	135 (69)	
Cord blood (N = 526)	203 (39)	69 (13)	55 (10)	79 (15)	323 (61)	
Donor type, n (% of aGVHD for each donor type)						.02
Sibling match	112	31 (28)	35 (31)	46 (41)		
Haploidentical	10	3 (30)	3 (30)	4 (40)		
Unrelated	60	11 (18)	10 (17)	39 (65)		
Cord blood	203	69 (34)	55 (27)	79 (39)		
Disease Risk Index, n (%)						<.01
Low	47 (12)	14 (12)	14 (14)	19 (11)	43 (6)	
Intermediate	244 (63)	67 (59)	71 (69)	106 (63)	285 (38)	
High	48 (12)	14 (12)	10 (10)	24 (14)	99 (13)	
Very high	6 (2)	3 (3)	0	3 (2)	13 (2)	
Nonmalignant	40 (10)	16 (14)	8 (8)	16 (10)	318 (42)	
Myeloablative conditioning, n (%)	178 (46)	55 (48)	50 (49)	73 (43)	382 (50)	.44
GVHD prophylaxis, n (%)						<.01
CsA-containing	4 (1)	2 (2)	1 (1)	1 (1)	28 (4)	
CsA/MMF *	285 (74)	81 (71)	84 (82)	120 (71)	479 (63)	
CsA/MTX	49 (13)	19 (17)	12 (12)	18 (11)	97 (13)	
Sirolimus/MMF	3 (1)	0	1 (1)	2 (2)	38 (5)	
T cell depleted	28 (7)	11 (10)	2 (2)	15 (9)	66 (9)	
Other	16 (4)	1 (1)	3 (3)	12 (7)	50 (7)	
Karnofsky Performance Status 90-100, n (%)	324 (84)	96 (84)	93 (90)	135 (80)	635 (84)	
Any infection, n (%)	360 (94)	106 (93)	97 (94)	157 (93)	657 (87)	<.01
Bacteria	293 (76)	81 (71)	76 (74)	136 (81)	518 (68)	<.01
Viruses	316 (82)	91 (80)	88 (85)	137 (82)	493 (65)	<.01
Fungi	56 (15)	11 (10)	13 (13)	32 (19)	63 (8)	<.01

CsA indicates cyclosporine A; MMF, mycophenolate mofetil; MTX, methotrexate.

* Mostly used for cord blood transplantation and reduced-intensity conditioning.

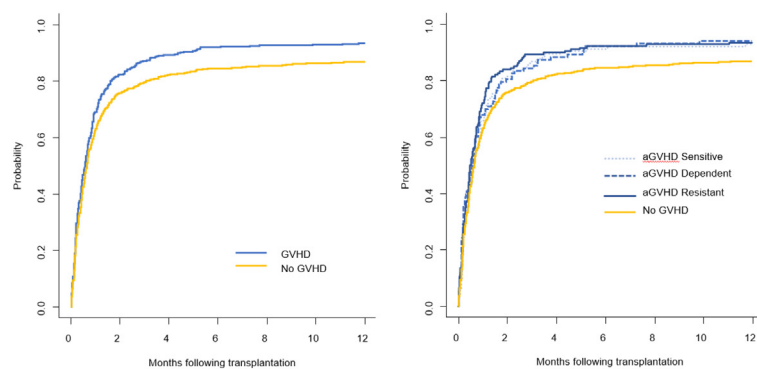
haploidentical donor transplantations, 13% of sibling donor transplantations, 15% of cord blood transplantations, and 20% of unrelated donor transplantations. Among transplantation recipients who developed any GVHD, SR GVHD accounted for ~40% (39% to 41%) of GVHD for sibling donor, haploidentical donor, and cord blood transplantations, but 65% of unrelated donor transplantations. Full demographic and transplantation characteristics for each GVHD cohort have been reported previously [21].

Cumulative Incidence

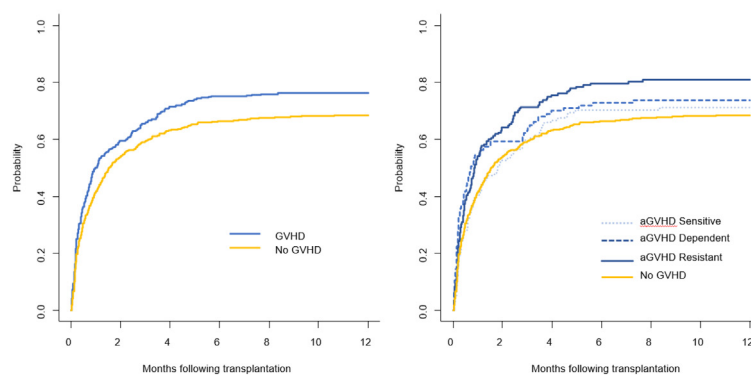
In the first year, 1017 patients (89%) developed at least 1 infection: 94% of those with aGVHD and 87% of the control group ($P < .01$) (Table 1). The CI of any infection by aGVHD subgroup was 93% for patients with SS aGVHD, 94% for those with SD aGVHD, and 93% for those with SR aGVHD (each $P < .01$ compared with no GVHD) (Figure 1A).

The CI of ≥ 1 bacterial infection was 76% for the patients with aGVHD, greater than the 68% for controls without GVHD

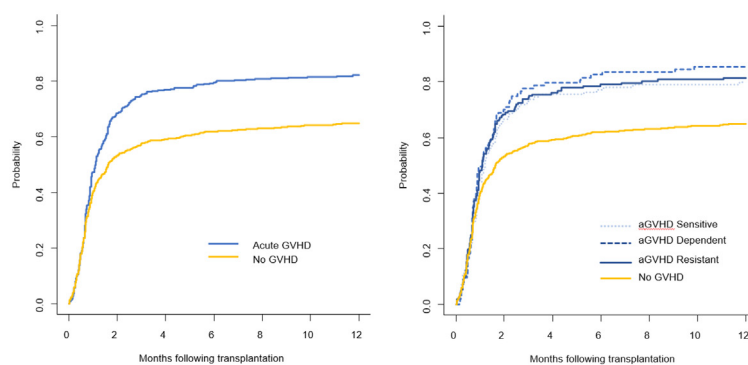
Panel A. CI of first infection is greater among patients with aGVHD than no GVHD (P-value <0.01).



Panel B. CI of first bacterial infection is greater among patients with aGVHD than no GVHD (P-value <0.01).



Panel C. CI of first viral infection is greater among patients with aGVHD than no GVHD (P-value <0.01).



Panel D. CI of first fungal infection is greater among patients with aGVHD than no GVHD (P-value <0.01).

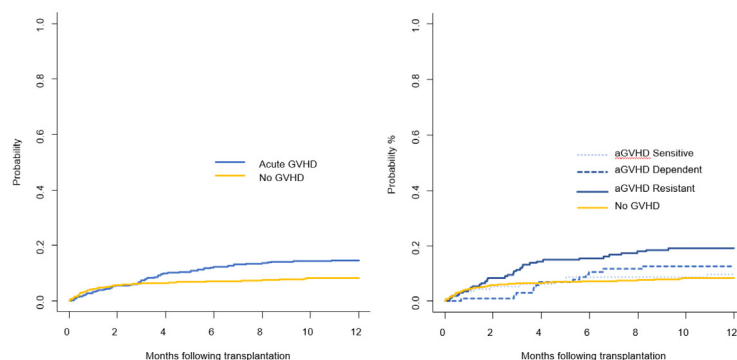


Figure 1. CI of first infection over the first year post-transplantation. GVHD patients are represented by progressive shades of blue. The reference group is no GVHD, in yellow. The right side of each panel shows the breakout of steroid response categories. (A) CI of first infection is greater in patients with aGVHD than in patients without GVHD ($P < .01$). (B) CI of first bacterial infection is greater in patients with aGVHD than in patients without GVHD ($P < .01$). (C) CI of first viral infection is greater in patients with aGVHD than in patients without GVHD ($P < .01$). (D) CI of first fungal infection is greater in patients with aGVHD than in patients without GVHD ($P < .01$).

($P < .01$) (Table 1). Viral infections developed in 82% of those with aGVHD versus 65% in controls ($P < .01$). The CI of fungal infections was 15% for those with aGVHD, compared with 8% in controls ($P < .01$) (Figure 1B).

Within each type of infection group (bacterial, viral, or fungal), SS aGVHD patients always had a lower percentage of patients with that type of infection compared with SD or SR aGVHD patients. The CI of bacterial infection in the first year after HCT was 76% overall, 71% for patients with SS aGVHD, 74% for SD aGVHD, and 81% for SR aGVHD, versus 68% for control patients without GVHD ($P < .01$). Of the 811 patients with a bacterial infection, the median time to the first bacterial event ranged from 11 to 23 days and varied by group.

The CI of viral infection in the first year post-transplantation was 82% overall, 80% for patients with SS aGVHD, 86% for SD aGVHD, and 82% for SR aGVHD versus 65% for patients without GVHD ($P < .01$) (Figure 1C). The median time to first viral event ranged from 24 to 28 days, depending on the group.

The CI of fungal infection in the first year post-transplantation was 10% overall, 10% for patients with SS aGVHD, 13% for SD aGVHD, and 19% for SR aGVHD versus 8% for patients without GVHD ($P < .01$) (Figure 1D). The median time to first fungal event was 53 days (range, 1 to 330 days).

Infection Episodes

After transplantation, 1017 of 1143 transplantation recipients had at least 1 infection, totaling 4561 distinct episodes of infection (Table 2). Patients with aGVHD averaged 4.9 infection

episodes in the first year post-transplantation, compared with 3.5 infection episodes in those without aGVHD. Among patients who had at least 1 infection, those with aGVHD averaged 5.3 infection episodes in the first year post-transplantation, compared with 4.0 infection episodes in those without aGVHD.

Of the 2476 bacterial infection episodes, 977 occurred among 293 aGVHD patients and 1499 occurred among 157 patients without GVHD. Gram-positive aerobic organisms accounted for more than two-thirds of bacterial infections (70% to 72% depending on subgroup), followed by Gram-negative aerobes (7% to 16%, with both the SR aGVHD and no GVHD groups with 16% of bacterial infections due to gram-negative anaerobes), *Clostridioides* (10% to 20%, with 20% of the bacterial infection episodes in SD aGVHD patients due to *Clostridioides*), mycobacteria (1% to 2%), filamentous bacteria (<1% to 1%), non-*Clostridioides* anaerobes (<1% to 1%), and rickettsia (0 to <1%). There were no infections identified with cell wall-deficient bacteria.

Of the 1940 viral infection episodes, 871 occurred among 316 aGVHD patients and 1069 occurred among 493 patients without GVHD. Herpesvirus organisms (DNA viruses) accounted for two-thirds of viral infection episodes (59% to 72% among subgroups), followed by non-herpes DNA virus infection episodes (13% to 27%), respiratory virus (RNA viruses) infection episodes (9% to 14%), nonrespiratory RNA virus infection episodes (0 to 2%), and hepatitis virus infection episodes (0 to <1%).

Among 140 fungal infection episodes, 65 occurred among 56 aGVHD patients and 75 occurred among 62 patients without GVHD. Yeast accounted for 23% of fungal infection episodes

Table 2
Infection Episodes in the First Year Post-Transplantation

Factor	Frequency					P Value
	All aGVHD	SS	SD	SR	No GVHD	
Patients per group	385	114	103	168	785	
All infection episodes, n (%)	1915	477	484	954	2646	<.01
Bacterial	977 (51)	224 (47)	232 (48)	521 (55)	1499 (57)	
Viral	871 (45)	239 (50)	237 (49)	395 (41)	1068 (40)	
Fungal	65 (3)	13 (3)	15 (3)	37 (4)	75 (3)	
Parasitic	2 (<1)	1 (<1)	0	1 (<1)	4 (<1)	
Bacterial group, n (%)						.07 for aGVHD vs no GVHD; .02 for aGVHD subsets vs no GVHD
Gram-positive aerobe	700 (72)	161 (72)	163 (70)	376 (72)	1048 (70)	
Gram-negative aerobe	119 (12)	21 (9)	17 (7)	81 (16)	237 (16)	
Filamentous (<i>Nocardia</i> , <i>Actinomyces</i>)	6 (1)	2 (1)	2 (1)	2 (<1)	6 (<1)	
<i>Clostridioides</i>	131 (13)	34 (15)	46 (20)	51 (10)	190 (13)	
Anaerobe	5 (1)	2 (1)	0	3 (1)	5 (<1)	
Rickettsia	1 (<1)	1 (<1)	0	0	3 (<1)	
Mycobacteria	15 (2)	3 (1)	4 (2)	8 (2)	10 (1)	
Viral group, n (%)						.18 for aGVHD vs no GVHD; <.01 for aGVHD subsets vs no GVHD
Herpesvirus	574 (66)	173 (72)	166 (70)	235 (59)	658 (62)	
Respiratory virus	101 (12)	34 (14)	22 (9)	45 (11)	133 (12)	
Other DNA virus (adenovirus, parvovirus, BK virus)	185 (21)	32 (13)	45 (19)	108 (27)	267 (25)	
Other RNA virus (enterovirus, rotavirus, norovirus)	11 (1)	0	4 (2)	7 (2)	9 (1)	
Hepatitis	0	0	0	0	1 (<1)	
Fungal group, n (%)						<.01
Yeast	15 (23)	6 (46)	0	9 (24)	46 (61)	
Mold	50 (77)	7 (54)	15 (100)	28 (76)	29 (39)	

among aGVHD patients and 61% of episodes in patients without GVHD, and molds accounted for 77% of fungal infection episodes among aGVHD patients and only 39% of episodes in patients without GVHD ($P < .01$). Comparing the incidence of yeast and mold infections is difficult, given that the 2-week interval for yeast infection episodes could lead to a higher number of events over a study risk period or over the year of the study compared with mold infections, which had a 12-week interval per episode.

Infection Density

Infection density for each assessment period was computed by examining the infection episodes reported during defined periods of risk. Given the similar median time to onset of aGVHD in all 3 groups (41 days in the SS group [range, 14 to 153 days], 37 days in the SD group [range, 11 to 166 days]), and slightly earlier onset of aGVHD in the SR cohort of 35 days [range, 10 to 170 days]), in the first analysis, we examined infection density according to the three classic infection risk periods associated with transplantation (Table 3, Figure 2).

In the first year post-transplantation, the infection density was 1.35 events per 100 patient-days at risk overall, 1.24 for patients with SS aGVHD ($P = .19$), 1.40 for SD aGVHD ($P < .01$), and 2.49 for SR aGVHD ($P < .01$), versus 1.16 for patients without aGVHD (control reference group) (Table 3). Between 31 and 100 days post-transplantation, SS aGVHD and SR aGVHD patients had more infections compared with the controls ($P = .03$ and $< .01$, respectively) (Figure 2).

The infection density of bacterial infections over the first year post-transplantation was .73 events per 100 patient-days at risk overall, .58 for patients with SS aGVHD ($P = .09$), .67 for

SD aGVHD ($P = .79$), and 1.36 for SR aGVHD ($P < .01$) versus .66 for patients without aGVHD (reference group). The density of bacterial bloodstream infections was highest during the day 0 to day +30 interval and decreased over time. In the early initial 30-day period post-transplantation, SD aGVHD and SR aGVHD patients had a higher density of bacterial infections, 2.95 and 3.03 infections per 100 patient-days at risk ($P = .04$ and $< .01$, respectively), compared with 2.33 infections per 100 patient-days at risk in the no GVHD group.

The infection density for viral infections over the first year post-transplantation was .57 events per 100 patient-days at risk overall, .62 for patients with SS aGVHD ($P < .01$), .69 for SD aGVHD ($P < .01$), and 1.03 for SR aGVHD ($P < .01$) versus .47 for patients without GVHD (reference group). In the early initial 30 days post-transplantation, patients who were eventually classified as SD aGVHD and SR aGVHD had a higher density of viral infections, 2.36 and 2.45 infections per 100 patient-days at risk ($P = .02$ and $< .01$ respectively), compared with 1.74 infections per 100 patient-days at risk in the no GVHD group.

The infection density for fungal infections over the first year post-transplantation was .041 events per 100 patient-days at risk overall, .034 for patients with SS aGVHD ($P = .93$), .043 for SD aGVHD ($P = .33$), and .096 for SR aGVHD ($P < .01$) versus .033 for patients without GVHD (reference group).

In the second analysis of infection density, including only the patients who developed aGVHD, time periods were based on landmarks experienced by the individual patients in this cohort (Figure 3). The reference group was patients classified as those with SR GVHD. Because the aGVHD classified patients had not required treatment for aGVHD prior to engraftment or

Table 3
Infection Density in the First Year Post-Transplantation

Factor	Frequency				Total
	aGVHD Treated with Systemic Steroids			No GVHD	
	SS	SD	SR		
Patients per group	114	103	168	785	1017
All infection episodes	477	484	955	2647	4536
Infection density from day 0 through 1 year (per 100 patient-days at risk)					
Any infection	1.24	1.4*	2.49*	1.16	1.35
Bacterial	.58	.67	1.36*	.66	.73
Viral	.62*	.69*	1.03*	.47	.57
Fungal	.034	.043	.096*	.033	.041
Infection density from day 0 through day +30 (per 100 patient-days at risk)					
Any infection	4.30	5.34 [†]	5.91 [†]	4.24	4.55
Bacterial	2.11	2.95*	3.03*	2.33	2.47
Viral	2.08	2.36 [‡]	2.45*	1.74	1.94
Fungal	.117	.032	.139	.16	.141
Infection density from day +31 through day +100 (per 100 patient-days at risk)					
Any infection	2.53 [§]	2.18	3.98*	2.15	2.45
Bacterial	1.05	.99	2.13*	1.23	1.31
Viral	1.45*	1.17 [†]	1.68*	.87	1.07
Fungal	.038	.028	.167*	.043	.058
Infection density from day +101 through 1 year (per 100 patient-days at risk)					
Any infection	.47	.67*	1.15*	.42	.52
Bacterial	.25	.29	.66*	.24	.29
Viral	.20	.33*	.44*	.16	.21
Fungal	.022	.049*	.056*	.012	.021

* $P < .01$.

[†] $P = .01$.

[‡] $P = .2$.

[§] $P = .03, .04$.

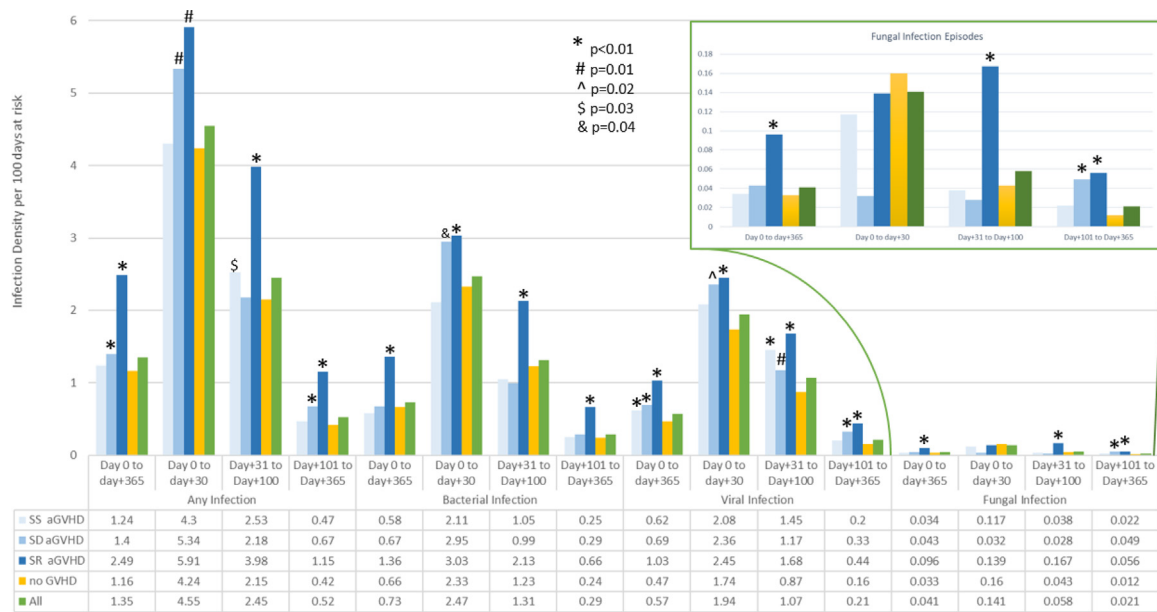


Figure 2. Density of infection episodes per 100 days at risk, using classic infection risk periods following transplantation. GVHD patients are represented by progressive shades of blue: light blue for SS, medium blue for SD, and dark blue for SR aGVHD treatment cohorts. The reference group is no GVHD, in yellow bars. Green bars summarize the entire study cohort. Figure 4 provides additional information concerning statistical comparisons.

prior to starting steroids for aGVHD treatment, there would be no expected overall difference in infection density between these 2 periods; however, some slight differences did occur. Patients eventually classified as SD had slightly more bacterial infections (density, 4.9 infections; $P = .04$) and slightly fewer

fungal infections (density, 0 infections; $P = .53$) prior to engraftment compared with patients who were eventually classified as SR (Figure 4). In the period from engraftment to the day prior to the start of aGVHD treatment, SD aGVHD patients had slightly fewer bacterial infections, possibly an

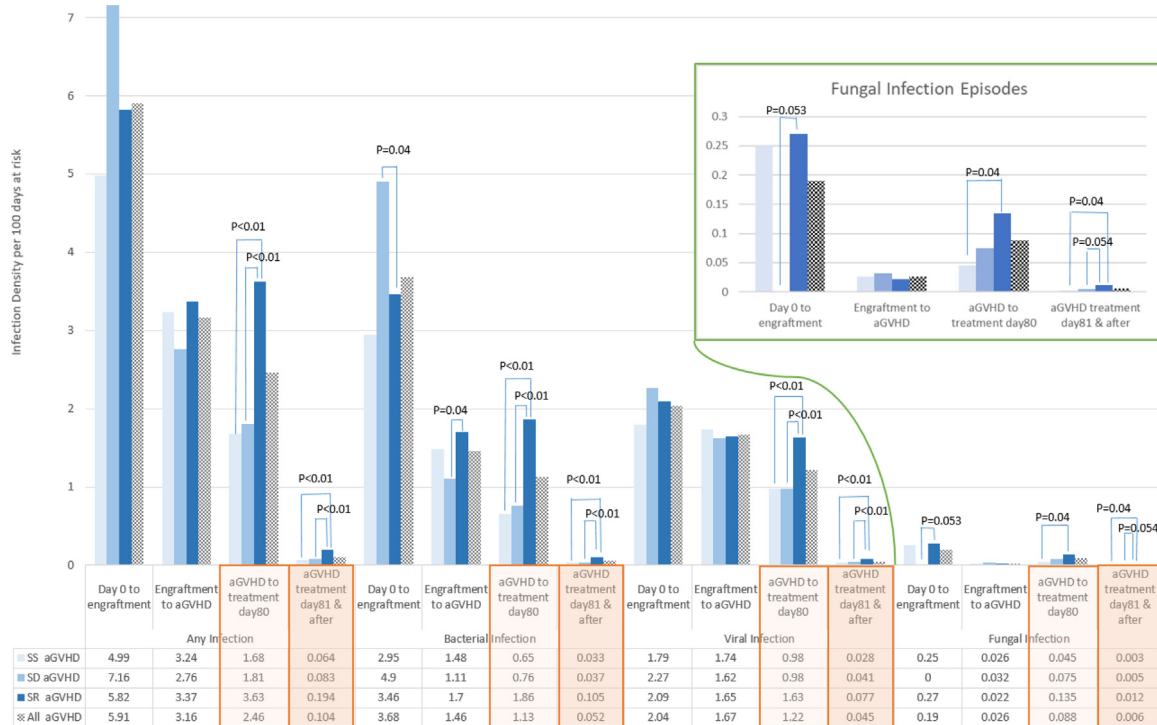


Figure 3. Density of infection episodes per 100 days at risk, specific to GVHD-derived risk periods following transplantation. Only patients who developed GVHD could have GVHD-specific risk periods defined and thus could be included in this figure. The reference group is SR GVHD. GVHD patients are represented by progressive shades of blue: light blue for SS, medium blue for SD, and dark blue for SR aGVHD treatment cohorts. Gray crosshatched bars summarize the aGVHD study cohort. Dark blue was chosen for the SR group to highlight the greater density of infection. Densities were recalculated with the start of each interval. Figure 4 provides additional information concerning statistical comparisons.

Comparison of aGVHD to no GVHD patients						Comparison of SS and SD aGVHD to SR aGVHD patients				
Classic definitions of infection risk periods						GVHD-specific infection risk periods				
		SS aGVHD	SD aGVHD	SR aGVHD	no GVHD			SS aGVHD	SD aGVHD	SR aGVHD
Any Infection	Day 0 to day+365	0.19	<0.01	<0.01	reference	Day 0 to day of engraftment Engraftment to pre GVHD treatment GVHD treatment to 80 days later 81 days GVHD treatment to 1 year	0.28 0.75 <0.01 <0.01	0.12 0.14 <0.01 <0.01	reference -- -- --	
	Day 0 to day+30	0.88	0.01	0.01	--					
	Day+31 to day+100	0.03	0.87	<0.01	--					
	Day+101 to day+365	0.17	<0.01	<0.01	--					
Bacterial Infection	Day 0 to day+365	0.09	0.79	<0.01	--	Day 0 to day of engraftment Engraftment to pre GVHD treatment GVHD treatment to 80 days later 81 days GVHD treatment to 1 year	0.40 0.44 <0.01 <0.01	0.04 (↑) 0.04 <0.01 <0.01	-- -- -- --	
	Day 0 to day+30	0.42	0.04	<0.01	--					
	Day+31 to day+100	0.18	0.08	<0.01	--					
	Day+101 to day+365	0.75	0.17	<0.01	--					
Viral Infection	Day 0 to day+365	<0.01	<0.01	<0.01	--	Day 0 to day of engraftment Engraftment to pre GVHD treatment GVHD treatment to 80 days later 81 days GVHD treatment to 1 year	0.50 0.77 <0.01 <0.01	0.73 0.91 <0.01 <0.01	-- -- -- --	
	Day 0 to day+30	0.17	0.02	<0.01	--					
	Day+31 to day+100	<0.01	0.01	<0.01	--					
	Day+101 to day+365	0.20	<0.01	<0.01	--					
Fungal Infection	Day 0 to day+365	0.93	0.33	<0.01	--	Day 0 to day of engraftment Engraftment to pre GVHD treatment GVHD treatment to 80 days later 81 days GVHD treatment to 1 year	0.87 0.92 0.04 0.04	0.053 (↑) 0.80 0.22 0.054	-- -- -- --	
	Day 0 to day+30	0.55	0.08	0.11	--					
	Day+31 to day+100	0.82	0.55	<0.01	--					
	Day+101 to day+365	0.16	<0.01	<0.01	--					

Figure 4. Heatmap of *P* values for transplantation time intervals examined for density of infection episodes. Orange shading (eg, <0.01) indicates more infections than in the reference group, and green shading (eg, <.01) indicates less infection than in the reference group. (Left) Comparison of aGVHD to no GVHD patients (reference), using listed infection risk periods following transplantation. (Right) Comparison of SS and SD aGVHD patients with SR aGVHD patients (reference) by GVHD-specific infection risk periods.

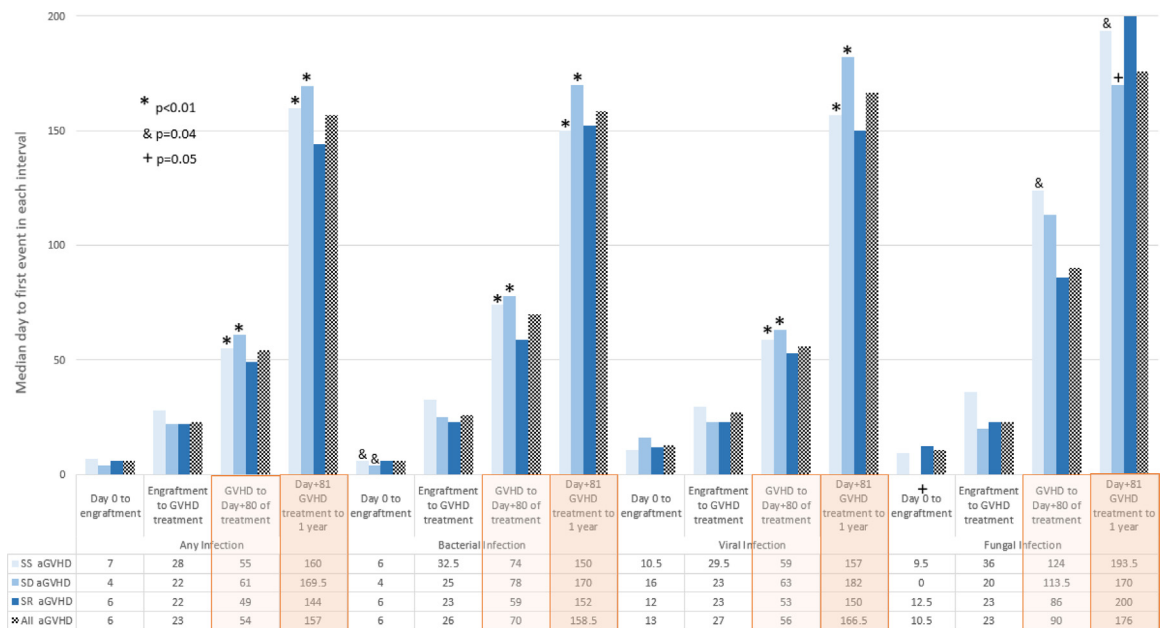


Figure 5. (A) CI of first infection in each infection risk period relative to the day of transplantation, using GVHD-specific infection risk periods: (1) day 0 to 1 day prior to engraftment; (2) engraftment to the start of GVHD treatment; (3) GVHD treatment to day +80 of treatment; and (4) GVHD treatment from day +81 to 1 year. Shown are the median days to first infection event in each interval, listed as the date of occurrence relative to the day of transplantation (day 0). GVHD patients are represented by progressive shades of blue: light blue for SS, medium blue for SD, and dark blue for SR aGVHD treatment cohorts. Gray crosshatched bars summarize the aGVHD study cohort. (B) CI of first infection in each risk period relative to start of each GVHD-specific infection risk interval: (1) day 0 to 1 day prior to engraftment; (2) engraftment to the start of GVHD treatment; (3) GVHD treatment to day +80 of treatment; and (4) GVHD treatment from day +81 to 1 year. The median days to first infection event in each interval are listed as the date of occurrence relative to the day of onset of each interval. GVHD patients are represented by progressive shades of blue: light blue for SS, medium blue for SD, and dark blue for SR aGVHD treatment cohorts. Gray crosshatched bars summarize the aGVHD study cohort.

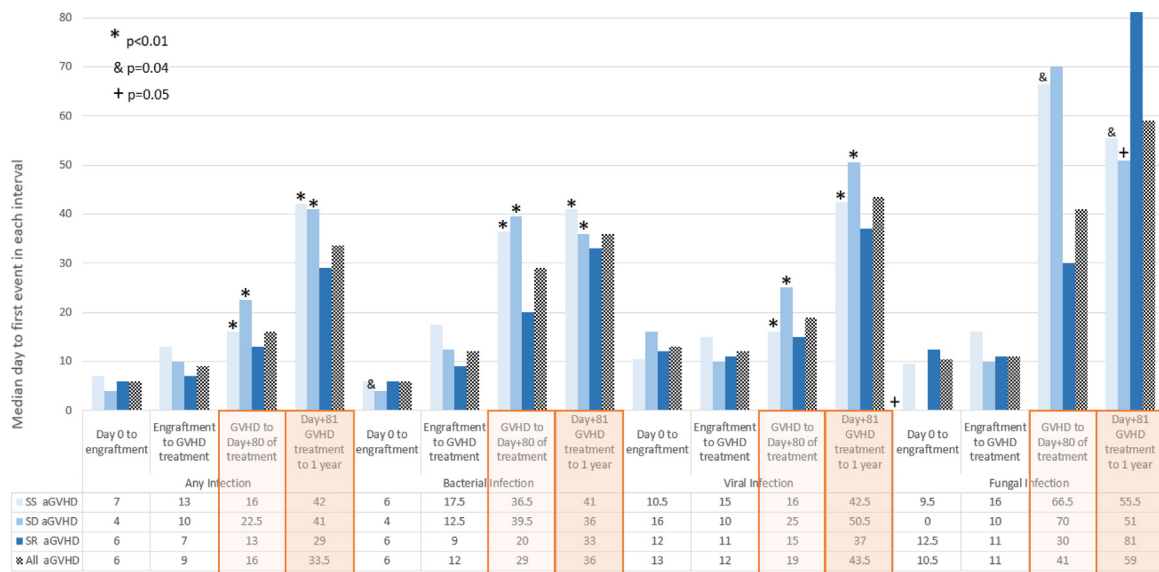


Figure 5. Continued.

artifact of their already being diagnosed and treated for bacterial infections prior to engraftment.

Infections in the next 2 risk periods, from the start of aGVHD treatment to 80 days of treatment, and then for the remainder of the first year post-transplantation, SS and SD patients had fewer bacterial and viral infections than SR patients. The effect was also present for fungal infections, but not as pronounced (Figure 4).

The median time of first infection event within each of these aGVHD patient-specific risk periods was similar among the SS, SD, and SR groups (Figure 5).

DISCUSSION

Our study provides an in depth look at CI and density of infections among patients who developed aGVHD and those who not develop aGVHD following allogeneic HCT. Infection density (infections per 100 patient-days at risk) was greater in patients treated for aGVHD compared with patients without GVHD (1.35 versus 1.16). Our study uniquely shows that the CI and density of infections for patients eventually classified as having SS aGVHD parallel those without GVHD (eg, the density difference of 1.24 versus 1.16 was not significant, at $P = .19$). We also demonstrate that steroid dependence and steroid resistance lead to more infections, with SR aGVHD patients having more infections at almost every time point and among infection organism groupings compared with either SS aGVHD or SD aGVHD. SR resistance had the highest density of infection (2.49), as well as a higher risk for 2-year overall mortality, as previously shown [21]. SD GVHD led to an intermediate number of infection episodes between the SS and SR groups. We interpret this as a meaningful increase in steroid-related infection morbidity for the SD GVHD patients, although without an increase in mortality, as shown previously [21].

We had hypothesized that infection episodes would be similar among patients in the 3 clinically encountered steroid response states in the initial 30 days post-transplantation or before engraftment, because patients would not be receiving high-dose steroid therapy for aGVHD during this month. However, patients in all 3 steroid response groups had more viral infection episodes in this period compared with controls, suggesting that the presymptomatic aGVHD syndrome is

permissive of this extra infectious burden either through mucosal/cutaneous barrier disruption or through another immune compromise. In addition, patients with SR aGVHD had more bacterial and fungal infections in the first month. It is acknowledged that post-transplantation infection frequency and severity can be a function of infection pretransplantation [1], but data obtained from this study more directly implicate aGVHD and its treatment sensitivity as a defined clinical risk factor.

Our study cohort was a large population of 385 patients with aGVHD. Besides including both adult and pediatric patients, there were >100 patients in each response group (114 with SS aGVHD, 103 with SD aGVHD, and 168 with SR aGVHD). Miller et al. [1] found that more total infections occurred after day +60 among 256 patients with aGVHD compared with 175 patients without aGVHD, with no review of steroid response categories. A 2017 study examining patterns of infection among 127 adults with SR aGVHD included no comparisons to SS aGVHD, SD aGVHD, or patients without GVHD [2]. For 198 children in a lookback study of bacteremias and fungal disease, aGVHD was graded as mild or severe but not classified by steroid response [4].

GVHD treated with systemic steroids is known to be associated with specific infection risks, such as drug-resistant bacteria [30]. Our study found that patients with aGVHD had 1.5 more infection episodes in the first year following transplantation compared with patients without GVHD, although this is a much lower figure than the 3- to 6-fold decrease reported by Miller et al. [1]. Chronic GVHD status was not assessed, since the end of immune suppression varied between acute and chronic GVHD, and thus the infection risk period was not directly tied to either GVHD syndrome. The specifics of infection in relation to the timing of individual organ involvement with GVHD are not available without additional review, and as such are beyond the scope of the available data.

The parent study for this analysis reviewed demographic data [21]. This detailed analysis of infections did not identify any other significant confounders of infection risk, such as relapse, ATG use, post-transplantation cytoxan use, or alternative treatments for GVHD. Although the incidence of GVHD was lower in haploidentical donor transplantations and

somewhat similar across the other donor sources, the risk of SR GVHD was notably higher in those receiving unrelated donor grafts and lower and similar in recipients of sibling donor, haploidentical donor, and cord blood grafts. Therefore, any graft source assessments are confounded by the variable rate of GVHD, which is at least partly attributable to the different GVHD prophylaxis regimens used with different graft sources. The incidence and severity of GVHD and its steroid resistance confound any assessment of the infection burden that is not attributable primarily to the donor and graft source. In particular, this unique dataset shows that infections following SR GVHD in cord blood recipients were similar to those observed in SR GVHD after transplantation with other donor types.

There are several limitations to the data presented in this report. The influence of pretransplantation infections was not assessed in this study, and we had incomplete information regarding infections in the pretransplantation period. Antimicrobial agent use was not monitored or recorded, because many patients initially had been treated in other centers. In addition, viral monitoring is rarely done pretransplantation, so patients could have had unrecognized low-level viral reactivations.

In our study, the stratification of risk for bacterial infections began following the onset of treatment for aGVHD. As reported previously, the development of a bacterial bloodstream infection with an enteric organism after the onset of acute gastrointestinal GVHD is a strong predictor of transplantation-related mortality [3]. Bacteremias, even late bacteremias, occur 3 to 6 times more frequently than fungal disease in children with aGVHD [4]. One potential mechanism behind this observation is intestinal barrier disruption, which could contribute to both bacteremia and the pathogenesis of aGVHD.

Antimicrobial prophylaxis with antiviral, antifungal, and anti-*Pneumocystis* medication is traditionally continued during aGVHD steroid treatment [31]. Not all centers continue antibacterial prophylaxis during aGVHD steroid treatment, however, both because patients are no longer neutropenic and because of the desire to preserve gut flora. These data suggest that the risk of infection follows the onset of GVHD therapy, and that aggressive surveillance and targeted antimicrobial prophylaxis, particularly for those with SR GVHD, remains important.

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